Sprint 1 – Week 2 Progress Report

SWENG 894 Capstone

MicroMatch

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1. System Overview and Context

1.1 System Scope

MicroMatch is a web-based micro-internship marketplace designed to connect students with short-term, skill-based professional opportunities. The system enables organizations to post clearly scoped micro-projects and allows students to discover and engage with professional experience opportunities.

Scope of Sprint 1 (US-1 → US-5)

Sprint 1 focuses on establishing the core marketplace foundation:

- User authentication and registration
- Role-based access control (RBAC)
- Student profile management
- Organization profile management
- Micro-internship listing creation
- Basic project browsing and search

These features establish the minimum viable marketplace infrastructure required for end-to-end interaction between students and organizations.

Out of Scope for Sprint 1 (Planned for Later Sprints)

- Application workflows (US-6 → US-9)
- Deliverable submission (US-10 → US-11)
- Project completion and feedback (US-12 → US-13)
- Administrative moderation (US-14)
- Audit logging (US-15)

These features will be incrementally implemented in subsequent sprints to maintain manageable scope and iterative delivery.

1.2 Week 2 Scope

Sprint 1 – Week 2 focused on implementing profile management and project listing functionality, building on the authentication foundation completed in Week 1.

Scope of Week 2 (US-2 → US-4)

- Student profile CRUD (Issue 31 & 33)
- Organization profile CRUD (Issue 32 & 41)
- Project listing creation (Issue 34 & 35)

These features allow authenticated users to manage their profiles and enable organizations to post micro-internship opportunities.

Out of Scope for Week 2

- Project browsing (Week 3)
- Frontend authentication flows (Week 3)
- End-to-end workflow validation (Week 3)

1.3 System Goals and Objectives

System Goals

- Allow students to create and update profiles
- Allow organizations to create and update company profiles
- Enable organizations to post micro-internship listings
- Maintain secure role-based access control using JWT

Alignment with Week 2 Work

Week 2 implemented SQLAlchemy models and secured API endpoints necessary for structured profile and listing management.

1.4 Functional Requirements to User Stories Mapping

Functional Requirement	User Story
FR-1: The system shall allow users to create an account.	US-1

FR-2: The system shall support role-based access control.	US-1
FR-3: The system shall allow students to create/edit profile.	US-2
FR-4: The system shall allow organizations to create profile.	US-3
FR-5: The system shall allow orgs to post listings.	US-4
FR-6: The system shall allow students to browse/search listings.	US-5
FR-7: The system shall allow students to apply.	US-6
FR-8: The system shall track application status.	US-8
FR-9: The system shall allow orgs to accept/reject.	US-9
FR-10: The system shall allow students to submit deliverables.	US-10
FR-11: The system shall allow orgs to review deliverables.	US-11
FR-12: The system shall allow marking projects complete.	US-12
FR-13: The system shall allow feedback submission.	US-13
FR-14: The system shall allow admin moderation.	US-14
FR-15: The system shall log system actions.	US-15

This traceability ensures alignment between formal IEEE-style functional requirements and Agile user stories used for sprint planning.

2. Sprint Planning

2.1 Groomed Product Backlog with Effort Estimation

ID	User Story	Priority	Story Points	Sprint	Rationale
US-1	Registration/Login	High	5	Sprint 1	Required for all access
US-2	Student Profile	High	5	Sprint 1	Core student data
US-3	Org Profile	High	5	Sprint 1	Required before posting
US-4	Post Listing	High	8	Sprint 1	Core marketplace feature
US-5	Browse/Search	High	5	Sprint 1	Enables discovery
US-6	Apply	High	5	Sprint 2	Core interaction
US-7	Review Applications	Medium	5	Sprint 2	Depends on applications
US-8	Track Status	Medium	3	Sprint 2	Enhances UX
US-9	Accept/Reject	Medium	3	Sprint 2	Workflow control
US-10	Submit Deliverables	Medium	5	Sprint 3	Completion flow
US-11	Review Deliverables	Medium	5	Sprint 3	Completion validation
US-12	Mark Complete	Medium	3	Sprint 3	Project closure
US-13	Feedback	Low	5	Sprint 3	Post-completion feature
US-14	Admin Moderation	Low	5	Sprint 3	Governance feature
US-15	Logging	Low	8	Sprint 3	Cross-cutting concern

Effort Estimation:

Story points follow a Fibonacci sequence, reflecting implementation complexity, dependency, and uncertainty.

2.2 Prioritization Rationale

Priority Definitions

- High: Core MVP functionality required for end-to-end workflow.
- Medium: Important workflow enhancements dependent on core features.
- Low: Administrative or cross-cutting features not required for initial MVP validation.

Sprint 1 focuses on High-priority items because they form the foundation of the platform's authentication and marketplace functionality. Without these core features, other dependent workflows cannot be implemented or tested, so completing them first ensures meaningful end-to-end functionality for the MVP.

2.3 Sprint 1 Timeline (3 Weeks)

Week 1 – Authentication Foundation (DONE)

- Environment setup
- Database schema implementation
- User model creation
- Password hashing
- JWT authentication
- Role-based access logic

Week 2 – Profile & Listing Features (DONE)

- Student profile CRUD
- Organization profile CRUD
- Listing creation

Week 3 – Marketplace Discovery (NEXT UP)

- Browse listings
- Basic search functionality
- End-to-end validation of Sprint 1 workflow

2.4 Week 2 Sprint Goal

Implement profile management and project listing functionality to extend the authenticated system into a structured marketplace backend.

2.5 Sprint 1 – Week 2 Backlog Summary

Total Story Points Committed: 18
Completed: 18
Remaining: 0

Sprint Item	Story Points	Related US	Reason Selected
Implement StudentProfile model	5	US-2	Core student data
Implement OrganizationProfile model	5	US-3	Required for posting
Implement Student Profile CRUD endpoints	5	US-2	Enable profile management
Implement Organization Profile CRUD endpoints	5	US-3	Enable org profile management
Implement Project Listing model	5	US-4	Data persistence for postings
Implement Project Creation endpoint	8	US-4	API for org to create listings

Capacity Reasoning:

Capacity was calculated based on ~15–20 effective development hours. Modeling and endpoint implementation were prioritized before UI integration.

All planned story points were completed within the estimated capacity, indicating realistic estimation and effective scope control.

2.6 Definition of Done (Week 2)

For Week 2 backlog items, the DoD includes:

- SQLAlchemy model implemented
- Foreign key constraints enforced

- Role validation confirmed
- CRUD endpoints functional
- Proper HTTP status codes returned
- Tested via Swagger/Postman
- Code committed and pushed to GitHub

2.5 Acceptance Criteria (Week 2)

US-2: Student Profile Management

Related FR: FR-3

Priority: High

Create Student Profile

AC-US2-01

- Given an authenticated user with role = Student
- When the user submits valid profile data to POST /student/profile
- Then
  - A StudentProfile record is created
  - The profile is linked to the authenticated user via foreign key
  - A 201 Created response is returned

Update Student Profile

AC-US2-02

- Given an authenticated student with an existing profile
- When the user submits updated data via PUT /student/profile
- Then
  - The profile information is updated in the database
  - The updated data is returned in the response
  - A 200 OK status is returned

Role Restriction Enforcement

AC-US2-03

- Given an authenticated user with role = Organization
- When the user attempts to access student profile endpoints
- Then

    - Access is denied
    - A 403 Forbidden response is returned
    - No profile data is created or modified

US-3: Organization Profile Management

Related FR: FR-4

Priority: High

Create Organization Profile

AC-US3-01

- Given an authenticated user with role = Organization
- When the user submits valid profile data to POST /org/profile
- Then
    - An OrganizationProfile record is created
    - The profile is linked to the authenticated user
    - A 201 Created response is returned

Update Organization Profile

AC-US3-02

- Given an authenticated organization with an existing profile
- When the user submits updated data via PUT /org/profile
- Then
    - The organization profile is updated
    - A 200 OK status is returned

Student Access Restriction

AC-US3-03

- Given an authenticated user with role = Student
- When the user attempts to access organization profile endpoints
- Then
    - Access is denied
    - A 403 Forbidden response is returned

US-4: Post Micro-Internship Listing

Related FR: FR-5

Priority: High

Create Project Listing

AC-US4-01

- Given an authenticated organization user
- When valid project data is submitted to POST /projects
- Then
  - A Project record is created
  - The project is linked to the organization via foreign key
  - Default status is set to "open"
  - A 201 Created response is returned

Missing Required Fields

AC-US4-02

- Given an authenticated organization user
- When required project fields (e.g., title or description) are missing
- Then
  - The request is rejected
  - A 422 Unprocessable Entity response is returned
  - No project record is created

Student Posting Restriction

AC-US4-03

- Given an authenticated user with role = Student
- When the user attempts to create a project
- Then
  - Access is denied
  - A 403 Forbidden response is returned

Test Case ID	User Story	Functional Requirement	Acceptance Criteria
TC-SP-01	US-2	FR-3	AC-US2-01
TC-SP-02	US-2	FR-3	AC-US2-02
TC-SP-03	US-2	FR-3	AC-US2-03
TC-OP-01	US-3	FR-4	AC-US3-01
TC-OP-02	US-3	FR-4	AC-US3-02
TC-OP-03	US-3	FR-4	AC-US3-03

TC-PJ-01	US-4	FR-5	AC-US4-01
TC-PJ-02	US-4	FR-5	AC-US4-02
TC-PJ-03	US-4	FR-5	AC-US4-03

Acceptance criteria follow the Given–When–Then convention to ensure testable, verifiable results.

3. Software Testing

3.1 Week 2 Testing Focus

Testing focused on:

- Model validation
- Foreign key enforcement
- Role-based access control
- Endpoint validation
- Error handling

Both Unit Tests (model validation) and System Tests (endpoint testing) were specified.

3.2 User Acceptance Test Specification (Partial)

Test Case ID	Related Issue	Type	Preconditions	Steps	Expected Result
TC-ES-01	25	Spike	Tools installed	Launch frontend and backend	Both services start successfully
TC-ES-02	25	Spike	Services running	Access local URLs	Frontend and backend accessible
TC-DS-03	26	Spike	Database running	Execute schema creation script	Tables created with correct constraints
TC-DS-04	26	Spike	Sample data inserted	Execute relational queries	Foreign key constraints enforced
TC-AU-01	28	System	User does not exist	Call registration endpoint with valid data	User created, password hashed

TC-AU-02	28	System	Existing user email	Call registration endpoint	Error returned, no duplicate record
TC-AU-03	29	System	Registered user exists	Call login endpoint with valid credentials	JWT token returned
TC-AU-04	29	System	Registered user exists	Call login with invalid password	Authentication rejected
TC-RBAC-01	30	System	Authenticated user	Access protected endpoint	Access allowed/denied based on role
TC-SP-01	31	Unit	Student authenticated	Create StudentProfile via POST	Profile saved, user_id linked
TC-SP-02	33	System	Student authenticated	GET /student/profile	Returns profile details, 200 OK
TC-SP-03	33	System	Student authenticated	PUT /student/profile	Profile updated successfully
TC-OP-01	32	Unit	Organization authenticated	Create OrganizationProfile	Profile saved, role validated
TC-OP-02	41	System	Org authenticated	GET /org/profile	Returns profile, 200 OK
TC-OP-03	41	System	Org authenticated	PUT /org/profile	Profile updated successfully
TC-PJ-01	34	Unit	Organization authenticated	Create Project model instance	Project saved with FK
TC-PJ-02	35	System	Organization authenticated	POST /projects	Project created, students denied access
TC-PJ-03	35	System	Organization authenticated	POST /projects with missing fields	Returns 422 error

3.3 Requirements-to-Test Case Traceability Matrix

Functional Requirement	User Story	Test Case IDs
FR-3: Student profile management	US-2	TC-SP-01 → TC-SP-03
FR-4: Organization profile management	US-3	TC-OP-01 → TC-OP-03
FR-5: Project listing creation	US-4	TC-PJ-01 → TC-PJ-03

4. Source Code Development

4.1 Summary of Contributions

- Implemented StudentProfile model
- Implemented OrganizationProfile model
- Implemented Project model
- Added CRUD endpoints for profiles
- Added Project creation endpoint
- Enforced JWT-based role validation
- Tested endpoints via Swagger

4.2 Repository Access and Version Control

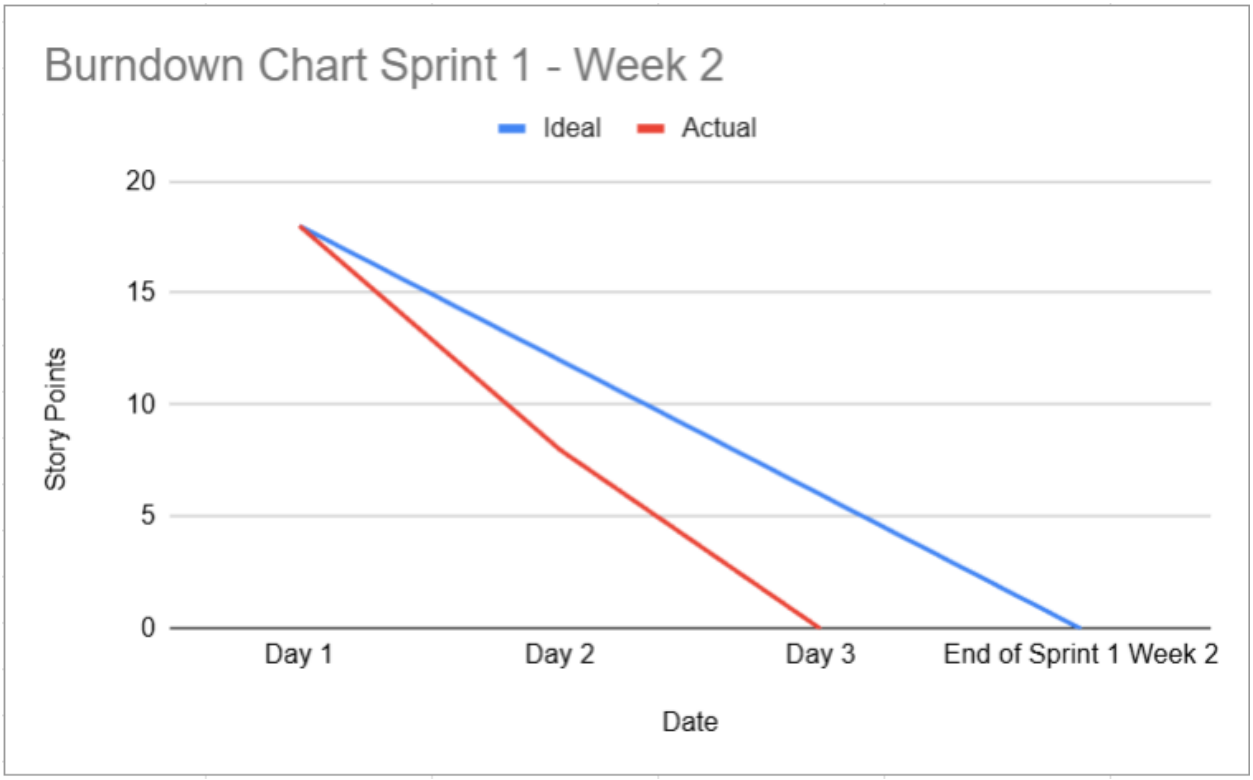
The repository is hosted on GitHub at:
https://github.com/chazdj/MicroMatch

All sprint artifacts and commits are visible. Access permissions have been verified.

4.3 Important Commits and Relation to Sprint Items

Commit	Related To	Purpose
Add test cases for US 1-auth	US-1	Confirms functionality
Implement StudentProfile model	US-2	Core student data entity
Implement OrganizationProfile model	US-3	Org profile entity
Add student profile CRUD endpoints	US-2	Enable student to manage profile
Add organization profile CRUD endpoints	US-3	Enable org to manage profile
Create Project model	US-4	Data persistence for postings
Implement Project Creation endpoint	US-4	API for org to create listings
Add test cases for US 2 -> US 4	US-2 → US-4	Confirms functionality

4.4 Burndown Chart Analysis



Total Story Points Committed: 18
Completed: 18
Remaining: 0

Week 2 focuses primarily on backend modeling and API endpoints; testing and frontend wireframes are partial.

5. Architectural Design Discussion

The system follows a layered architecture:

- **Presentation Layer**
  - React frontend responsible for UI rendering and interaction.
- **Application Layer**
  - FastAPI backend responsible for authentication, business logic, and workflow enforcement.
- **Data Layer**
  - PostgreSQL database maintaining persistent data with referential integrity.

Architectural Rationale

- Separation of concerns enhances maintainability.
- JWT-based authentication enables stateless scalability.
- Role-based access control enforces security boundaries.
- Layered architecture supports incremental Agile development.

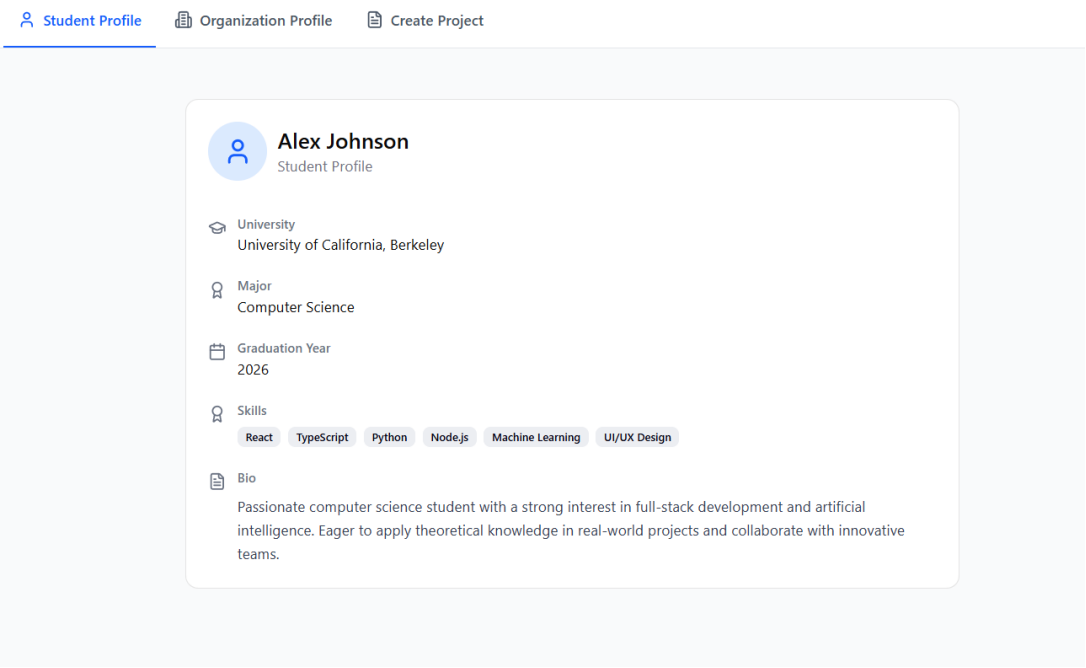
Decisions from Week 1 remain foundational (JWT, RBAC). Week 2 extends these to profile and project management.

6. UI Design

Wireframes were created for US-2, US-3, and US-4.

Each interface directly maps to backend models and enforces role restrictions.

6.1 Student Profile Page (US-2)



Description and Design Rationale

The Student Profile interface allows authenticated student users to create and manage their professional identity within the MicroMatch platform. This page directly supports:

- **US-2: Student Profile Management**
- Related Issues: #31 (Model), #33 (CRUD Endpoints)
- Functional Requirement: FR-3

Layout Structure

The interface is structured into three logical sections:

1. **Academic Information**
    a. University (text input)
    b. Major (text input)
    c. Graduation Year (numeric input)
2. **Skills Section**
    a. Skills (comma-separated or tag-style input)
3. **Professional Summary**
    a. Bio (multi-line text area)

This grouping reduces cognitive load by organizing related information together and follows basic usability heuristics of consistency and clarity.

Usability Considerations

- **Clear Labeling:** Each field is explicitly labeled to reduce ambiguity.
- **Input Constraints:** Graduation year restricted to numeric input to prevent invalid data.
- **Multi-line Bio Field:** Allows expressive input while maintaining structure.
- **Edit Mode vs View Mode:** The interface visually distinguishes between viewing and editing states to prevent accidental modification.
- **Save Confirmation:** Users receive visual confirmation after successful update.

Validation & Error Handling

- Required fields: University, Major
- Graduation year must be within a reasonable range
- Skills input sanitized to prevent injection

Error messages are displayed inline beneath fields to reduce friction.

Security Alignment

- Page is accessible only to users with role = Student.
- JWT authentication required.
- Users may only modify their own profile (ownership validation).

Alignment With Backend Design

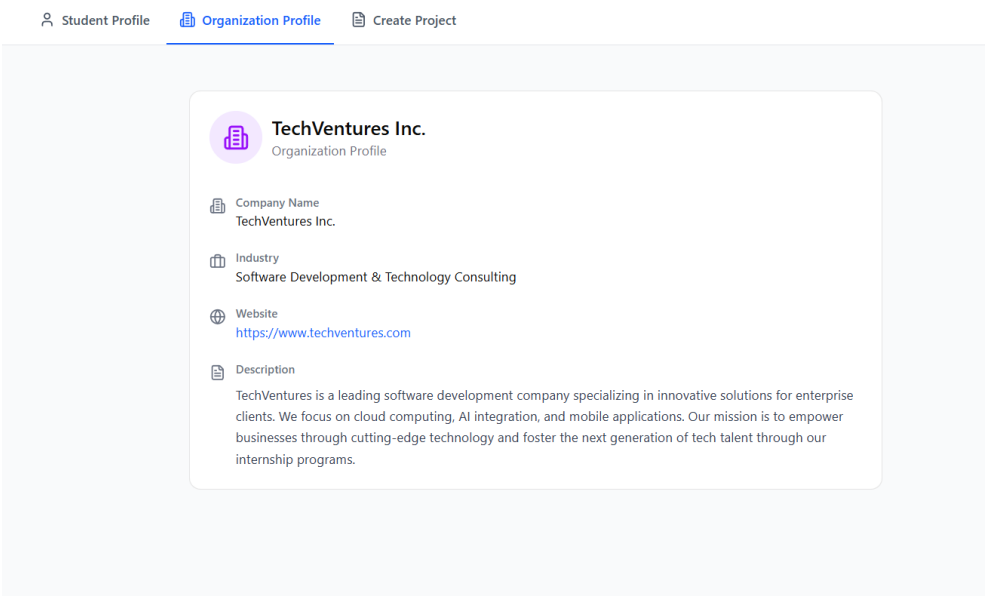
This UI maps directly to:

StudentProfile model fields:

- university
- major
- graduation_year
- skills
- bio

The wireframe reflects the one-to-one relationship between User and StudentProfile.

6.2 Organization Profile Page (US-3)



Description and Design Rationale

The Organization Profile page allows authenticated organization users to manage company information displayed to students. It supports:

- **US-3: Organization Profile Management**
- Related Issues: #32 (Model), #41 (CRUD)
- Functional Requirement: FR-4

Layout Structure

The page is organized into:

1. **Company Information**
    a. Company Name (text input)
    b. Industry (dropdown or text input)
    c. Website (URL input)
2. **Company Overview**
    a. Description (multi-line text area)

This layout emphasizes clarity and credibility by structuring public-facing information in a professional format.

Usability Considerations

- **URL Validation:** Website field ensures valid URL format.
- **Industry Dropdown (Optional):** Improves data consistency and future filtering.
- **Character Counter for Description:** Encourages concise but informative company summaries.
- **Preview Mode (Future Consideration):** Allows organization to preview how students will see their profile.

Validation & Error Handling

- Company Name required.
- Website validated for correct format.
- Description length capped to maintain readability.

Inline error messaging ensures correction without full page reload.

Security Alignment

- Accessible only to role = Organization.
- Students cannot access or modify this page.
- Ownership validation ensures organization edits only its own profile.

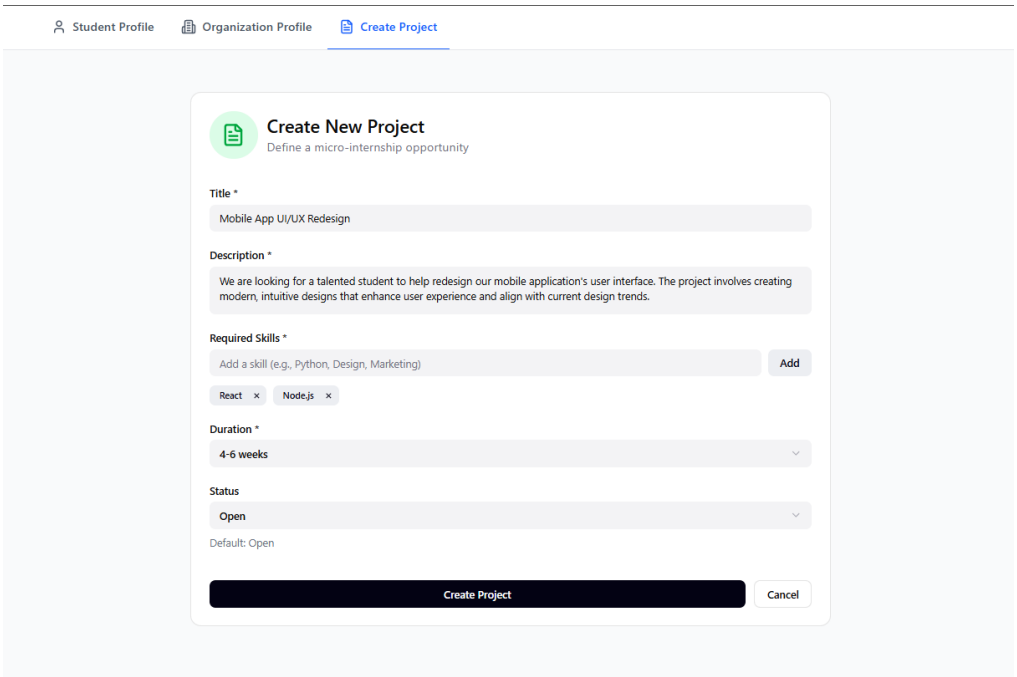
Alignment With Backend Design

Maps directly to OrganizationProfile model:

- company_name
- industry
- website
- description

This supports the foreign key relationship to User and enforces role-based constraints.

6.3 Create Project Page (US-4)



Description and Design Rationale

The Create Project page allows organizations to post structured micro-internship listings. It supports:

- **US-4: Post Micro-Internship Listing**
- Related Issues: #32 (Project Model), #35 (Endpoint)
- Functional Requirement: FR-5

Layout Structure

The interface is divided into:

1. **Project Basics**
    a. Title (text input)
    b. Duration (text or numeric input)
2. **Project Details**
    a. Description (multi-line text area)
    b. Required Skills (tag input)
3. **Status**
    a. Default set to "Open"
    b. Optional toggle (Open / Closed)

Usability Considerations

- **Clear Scoping:** Title and description fields are visually distinct to encourage structured listings.
- **Skill Tags:** Improves readability and prepares for search filtering in Week 3.
- **Default Status = Open:** Reduces user friction.
- **Confirmation Modal Before Submission:** Prevents accidental posting.

Validation & Error Handling

- Title required.
- Description required.
- Required Skills cannot be empty.
- Duration must follow defined format.

Backend returns appropriate HTTP codes (201, 422, 403).

Security Alignment

- Only role = Organization can access.
- Students receive 403 Forbidden.
- JWT required.
- Project linked via foreign key to OrganizationProfile.

Alignment With Backend Design

Maps directly to Project model:

- organization_id (FK)
- title
- description
- required_skills
- duration
- status
- created_at

The wireframe ensures all required backend fields are represented in the UI.

7. Backlog Grooming

No changes were made to the Sprint 1 backlog during Week 2.

All sprint items remain aligned with the originally defined MVP scope.

8. Conclusion

Week 2 successfully expanded the MicroMatch backend from authentication only to a structured marketplace foundation. Profile management and project listing functionality are now implemented and secured with role-based access control. The system is positioned for Week 3, which will focus on browsing, search functionality, and full workflow validation.